

LESSON 5-3 • SESSION 1 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Area of Triangles

Builds on Lesson 5-2 Session 1 • 6.G.A.1

TODAY'S LESSON

Each end piece of the trapezoid is a triangle. You find its area with $\frac{1}{2} \times b \times h$.

YOU NEED TO

Find a triangle's area, like $\frac{1}{2} \times 4 \times 6 = 12$. **You will use it in Practice Form A, problems 2 and 8.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** Find the **base** and the height.
- 2** **Multiply** base $\times$ height.
- 3** **Halve** the product.

WATCH ME*the steps in action***A triangle has base 5 m and height 10 m. Find its area.****Step 1** $b = 5, h = 10$.**Step 2** $5 \times 10 = 50$.**Step 3** $50 \div 2 = 25$.**Answer** **25 m²****TRY IT WITH HELP***fill in each step*

- 1.** A triangle has base 4 ft and height 6 ft. Find its area.

Step 1 $b = \underline{\hspace{2cm}}, h = \underline{\hspace{2cm}}$ **Step 2** $4 \times 6 = \underline{\hspace{2cm}}$ **Step 3** $24 \div 2 = \underline{\hspace{2cm}} \text{ ft}^2$

- 2.** A triangle has base 6 in and height 16 in. Find its area.

Step 1 $b = \underline{\hspace{2cm}}, h = \underline{\hspace{2cm}}$ **Step 2** $6 \times 16 = \underline{\hspace{2cm}}$ **Step 3** $96 \div 2 = \underline{\hspace{2cm}} \text{ in}^2$ **YOUR TURN***use the steps on your own*

- 3.** Find the area of each triangle.

a. $b = 8, h = 5$ _____**b.** $b = 3, h = 10$ _____**c.** $b = 7, h = 4$ _____**d.** $b = 9, h = 6$ _____**e.** $b = 12, h = 5$ _____**f.** $b = 10, h = 13$ _____

Identify Base and Height

Builds on Lesson 5-1 Session 1 • 6.G.A.1

TODAY'S LESSON

Every piece you cut from a trapezoid deck uses the same height, like 6 ft.

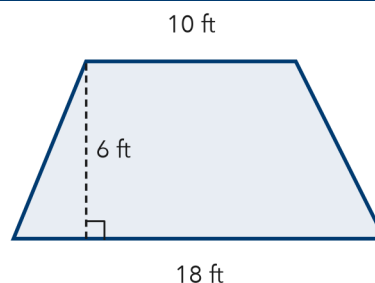
YOU NEED TOPick the height: the distance that makes a square corner, never a slanted side. **You will use it in Practice Form A, problems 3 and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Pick a side as the **base**.
- 2** Find the segment that makes a **square corner** with it.
- 3** That segment is the **height**. Skip slanted sides.

WATCH ME*the steps in action*

Bases 10 ft and 18 ft, slanted sides about 7 ft, dashed line 6 ft. Which is the height?

- Step 1** The base is the 18 ft side.
- Step 2** The 6 ft line makes a square corner.
- Step 3** The height is 6 ft, not the 7 ft slanted side.
- Answer** $h = 6$ ft

**TRY IT WITH HELP***fill in each step*

- 4.** A triangle has base 12 cm, slanted side 10 cm, dashed line 8 cm. Find the height.

- Step 1** The base is _____ cm.
- Step 2** The _____ cm line makes a square corner.
- Step 3** The height is _____ cm.

- 5.** Bases 9 in and 15 in, slanted sides 5 in, dashed line 4 in. Find the height.

- Step 1** The long base is _____ in.
- Step 2** The _____ in line makes a square corner.
- Step 3** The height is _____ in.

YOUR TURN*use the steps on your own*

- 6.** Write the height. "Corner" is the line that makes a square corner.

- a.** base 10, slanted 6, corner 5 _____ **b.** base 14, corner 9, slanted 11 _____
- c.** slanted 13, base 20, corner 12 _____ **d.** base 8, corner 3, slanted 5 _____

READY CHECK*a problem like the one in today's lesson*

- 7.** Bases 8 m and 18 m, slanted sides 13 m, height 12 m. Which number is h ?

- A** 8 m **B** 12 m **C** 13 m **D** 18 m

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____